

Mapping — and Stress-Testing — the SARS-CoV-2 Host Interactome

A live-data reproduction of Gordon et al.'s host-directed drug-repurposing map, extended with pathway enrichment and guilt-by-association prediction, then subjected to fourteen adversarial tests — including checks against an independent lab's dataset, an independent curated network, and an unbiased Y2H interactome — to find out which claims actually survive.

Source study Gordon et al., *Nature* 583:459–468 (2020), doi:10.1038/s41586-020-2286-9

Interactions analyzed 332 **Viral proteins** 27 **Validation tests** 14

ABSTRACT

Gordon et al. (2020) used affinity-purification mass spectrometry to identify 332 high-confidence physical interactions between 27 SARS-CoV-2 viral proteins and human host proteins, then mapped 69 of those host proteins onto existing drugs as host-directed antiviral candidates. This report pulls that same interaction set live from its public NDEx record and layers three additional analyses on top: **(1)** pathway enrichment (GO/KEGG/Reactome via Enrichr) to characterize which biological processes each viral protein hijacks, **(2)** guilt-by-association network expansion (STRING) to predict host proteins that plausibly interact with a given viral protein but were not detected in the original screen, and **(3)** a fourteen-part adversarial validation suite built specifically to find out whether (2) produces real signal or database artifacts.

The enrichment step recovers biology consistent with the original paper — centrosome and mitotic pathways concentrated on Nsp13, DNA-replication pathways on Nsp1, mitochondrial-import pathways on Nsp4/Orf9b/Orf9c — though a coverage audit shows Reactome annotates only 68.7% of the 332 proteins, unevenly across viral proteins, so

cross-bait comparisons are not on equal footing. The prediction step produced two headline candidates: **PLK1** as a missing Nsp13 interactor, and **TOMM70** as a missing Nsp4 interactor. Validation shows an important asymmetry: the Nsp13–PLK1 call is specific (not a generic network hub; 0/25 hits under a degree-preserving null), reasonably robust across a parameter sweep, and has above-null but modest predictive value overall (11.1% recall / 3.5% precision on held-out real interactions, versus 0%/0% for the null) — but it has *not* been independently confirmed in any other published SARS-CoV-2 interactome. The Nsp4–TOMM70 call, originally framed as cross-validated, does not survive scrutiny as well: it is threshold-fragile (present only at the loosest STRING confidence setting) and every independent confirmation of TOMM70 as a real SARS-CoV-2 target (seven papers) implicates ORF9b, never Nsp4. **The most consequential finding came from following those next steps.** Re-running the prediction against a genuinely independent dataset (Stukalov et al. 2021, a separate lab and pipeline) as the answer key — rather than an internal split of Gordon's own data — collapsed mean recall from 11.1% to 0.07%, indistinguishable from a matched null. A follow-up diagnosis shows why: the method only works where a bait's interactors already form a tight STRING cluster, and that clustering is driven by literature co-mention (text-mining channel, $\rho +0.78$) rather than hard experimental evidence ($\rho +0.50$). In plain terms, the method predicts well between already-famous proteins and fails on novel biology. **Two closing tests split the headline calls apart.** Rebuilt on IntAct's curated *physical*-interaction graph — a structurally different network with no text-mining channel — Nsp13–PLK1 *reproduces* (6 of 40 Nsp13 preys are direct physical partners of PLK1, $p=0.0005$; PLK1 in the top $\sim 1.2\%$ of candidates, among other centrosome proteins), so it is not a STRING artifact; Nsp4–TOMM70 *does not* (0/8, unranked) and is withdrawn. The bottom line: Nsp13–PLK1 is a specific, cross-method-reproducible hypothesis about the machinery Nsp13 engages, still awaiting a direct binding assay; Nsp4–TOMM70 does not survive; and the prediction method as a whole should be treated as hypothesis-generating, not confirmatory. A final search for a bias-free confirmation substrate (HuRI, an unbiased systematic-Y2H interactome) maps the ceiling of what computation can settle here: HuRI is structurally blind to the centrosome (10 of 14 core proteins absent, PLK1 at degree 1), so no unbiased binary dataset can adjudicate Nsp13–PLK1 — every dataset that covers this biology is literature-curated.

Confirming it now strictly requires a wet-lab co-IP, and saying so plainly is more useful than manufacturing closure.

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FIGURE WALKTHROUGH — 1 OF 8

What pathways does the whole interactome point to?

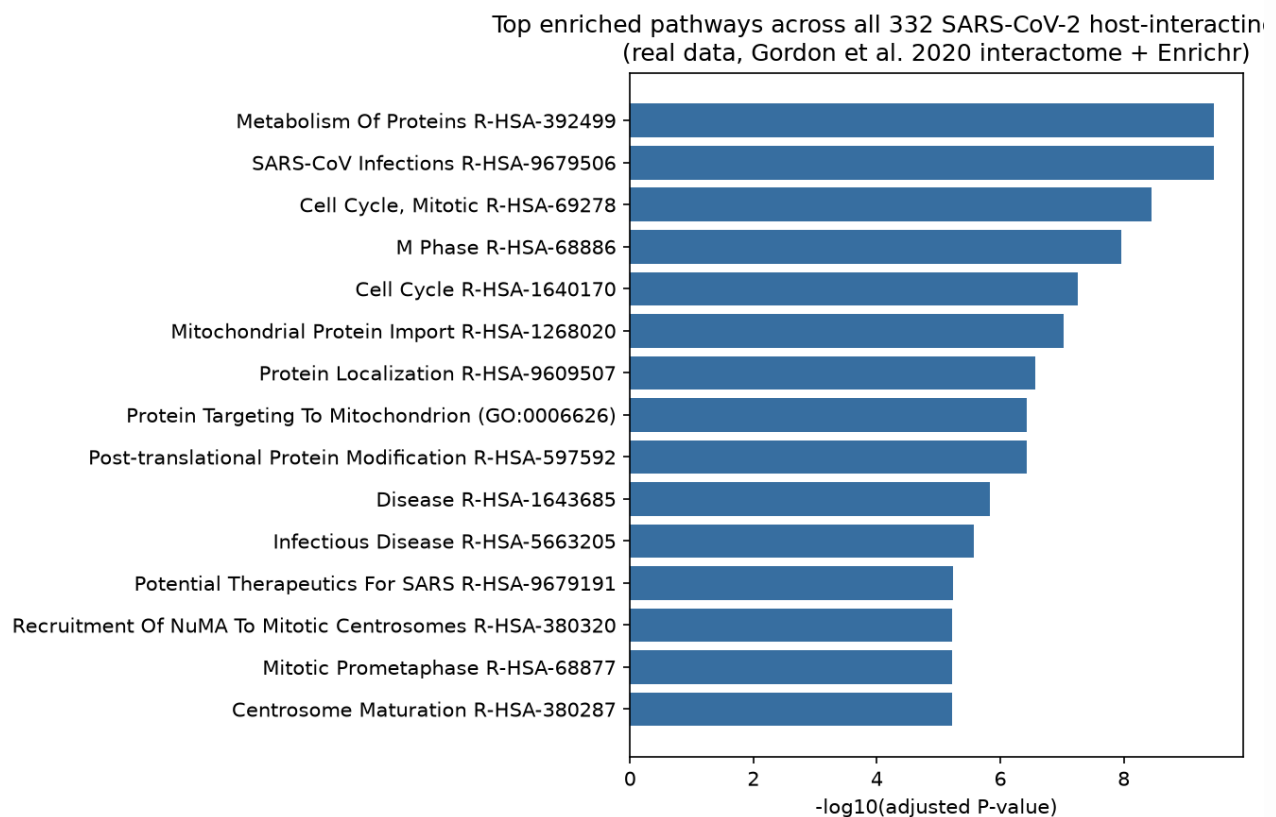


FIGURE 1

Top enriched pathways across all 332 host-interacting proteins

WHAT IT SHOWS

Each of the 332 human proteins bound by a SARS-CoV-2 viral protein was tested for statistical over-representation against three pathway databases (GO Biological Process, KEGG, Reactome, queried via Enrichr). The 15 most significant terms overall are plotted; bar length is $-\log_{10}(\text{adjusted p-value})$, so a longer bar means the 332-protein list is more improbably concentrated in that pathway than chance would predict.

HOW TO READ IT

This is a ranking, not a magnitude scale you should over-interpret bar-to-bar — treat it as "which processes does this virus concentrate its host contacts in," not "process X is N times more important than process Y."

WHAT'S ESTABLISHED VS. WHAT TO WATCH

Reactome's own *SARS-CoV Infections* and *Potential Therapeutics For SARS* terms appear near the top — these are curated directly from this literature, so their presence is expected and not independent confirmation (see Figure 5, which removes them). The remaining terms — Cell Cycle/M Phase, Mitochondrial Protein Import, Protein Targeting To

Mitochondrion — are not literature-circular and line up with the original paper's Fig. 2b clusters.

RESEARCH THIS YOURSELF

The full ranked list (240 significant terms) is in `data/pathways_overall.csv`. The coverage audit (`data/validation/coverage_audit.csv`) found Reactome annotates only 228/332 proteins (68.7%) and KEGG only 154/332 (46.4%) — before trusting this ranking as "the biology," check whether your protein of interest is even in the library's universe.

FIGURE WALKTHROUGH — 2 OF 8

Which viral protein drives which pathway?

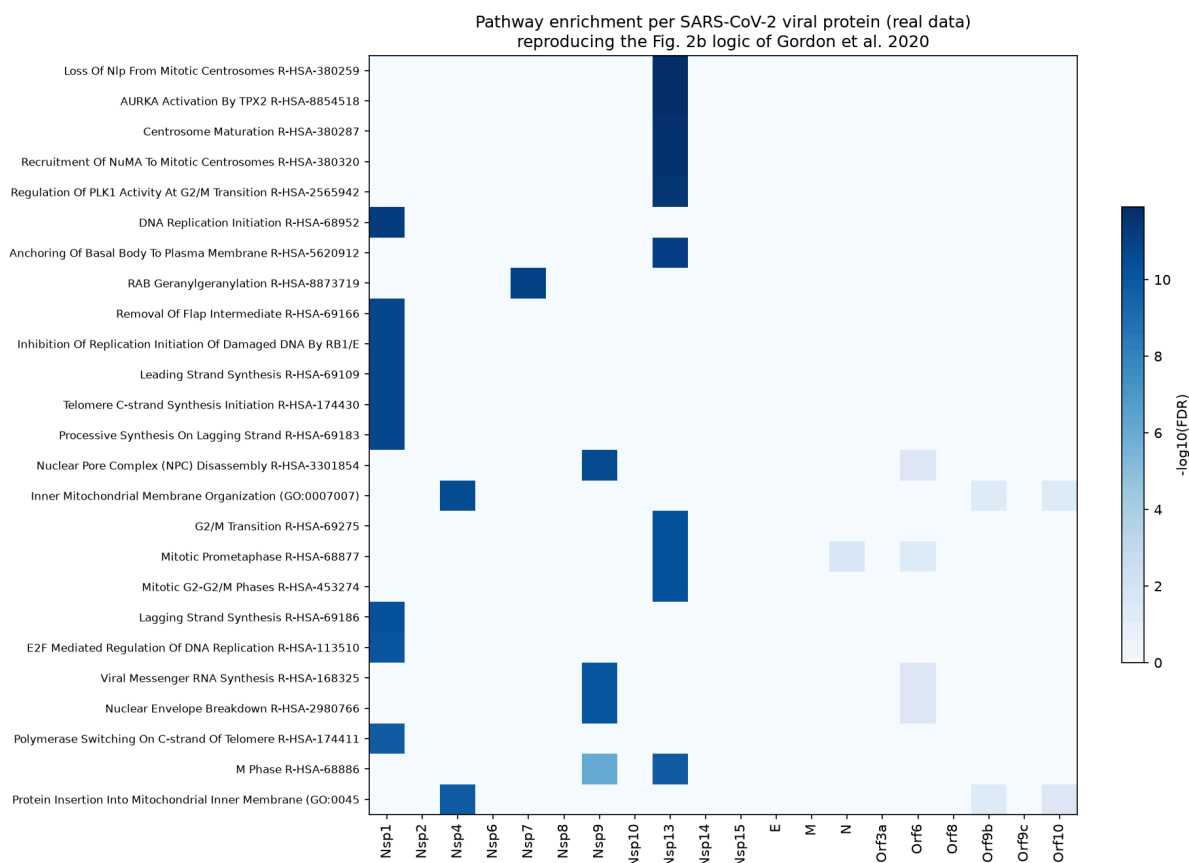


FIGURE 2

Pathway enrichment broken out per viral protein

WHAT IT SHOWS

Rows are the 25 pathway terms that reach their strongest significance somewhere in the dataset; columns are the 17 (of 27) viral proteins with enough known host interactors (≥ 3) to run enrichment on their own. Cell shading is $-\log_{10}(\text{FDR})$ for that specific pathway within that specific bait's prey set — dark means that viral protein's targets are themselves concentrated in that pathway, not just the dataset as a whole.

HOW TO READ IT

Look for vertical concentration. **Nsp13**'s column is dark across nearly every centrosome/mitotic row (*AURKA Activation By TPX2, Centrosome Maturation, Recruitment Of NuMA To Mitotic Centrosomes*) — this is the biological basis for the PLK1 prediction in Figure 3. **Nsp1**'s column is dark across DNA-replication rows (*Leading/Lagging Strand Synthesis, Telomere C-strand Synthesis*). **Nsp9** shows nuclear-envelope/M-phase terms.

WHAT'S ESTABLISHED VS. WHAT TO WATCH

Many columns are almost entirely blank (Nsp2, Nsp6, E, M, Orf3a, Orf8, Orf9c). Do not read this as "these viral proteins do less" — the coverage audit found annotation depth varies systematically by bait (Nsp12, Orf7a, Nsp2, M, and Orf3a average the lowest library coverage per gene), so a blank column can mean "under-annotated," not "biologically inert."

RESEARCH THIS YOURSELF

Pull any cell's underlying gene list from `data/pathways_per_bait.csv` (filtered by `query = bait name, Term = pathway`) and cross-check it against Reactome's own pathway diagram browser at reactome.org.

The strongest surviving prediction: Nsp13 → PLK1

Nsp13 centrosome network: PLK1 predicted as missing component
(guilt-by-association via STRING, 15 edges to known Nsp13 preys)

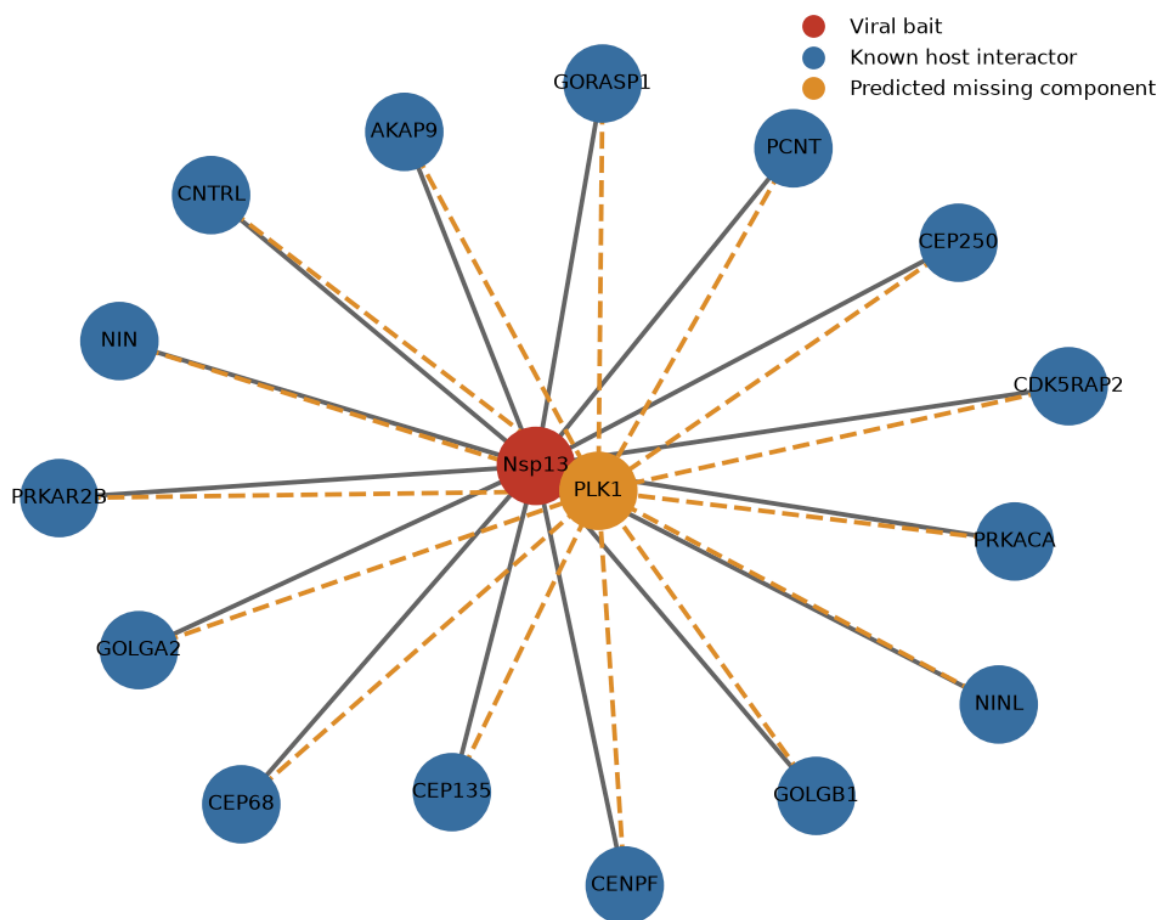


FIGURE 3

Nsp13's centrosome network, with PLK1 as a predicted missing component

WHAT IT SHOWS

● **Red** — Nsp13, the viral bait. ● **Blue** — 14 of Nsp13's real, mass-spectrometry-confirmed host interactors from the original 332 (e.g. **CEP135**, **PCNT**, **CNTRL**, **CENPF**, **AKAP9**), all centrosome-complex proteins. ● **Orange** — **PLK1**, not present anywhere in the original 332 interactions, surfaced by asking STRING's network-expansion algorithm which protein is best-connected to this whole known set. Solid grey edges are real (AP-MS observed); dashed orange edges are the predicted, unobserved links.

HOW TO READ IT

PLK1 (Polo-like kinase 1) sits at a hub connected to nearly every blue node — meaning, per STRING's combined evidence (co-expression, curated pathway membership, physical experiments, text-mining), it is tightly co-functional with the exact centrosome machinery Nsp13 is already known to bind. PLK1 is centrosome biology's master switch: it drives centrosome maturation and G2/M-transition licensing, the same processes Figure 2 shows Nsp13 already dominates.

WHAT'S ESTABLISHED VS. WHAT TO WATCH

This call passed every internal check: it is not a generic promiscuous hub (0 of 25 hits when the same-size gene set was replaced with random proteins from the pool), it never appeared for unrelated viral proteins (E, M, Orf7a) used as negative controls, and it survived 5 of 6 settings in a confidence-threshold sweep. Held-out testing recovered 27.5% of Nsp13's hidden real interactions on average — one of the two best scores in the dataset. **But:** querying IntAct (an independent interaction database aggregating multiple separate SARS-CoV-2 interactome papers) turns up zero record of PLK1 binding Nsp13. PLK1's only documented SARS-CoV-2 partner elsewhere is the Spike protein (Cattin-Ortolá et al. 2021, PMID 34504087) — a different protein entirely.

RESEARCH THIS YOURSELF

Look up PLK1 (UniProt P53350) and Nsp13's role as SARS-CoV-2 helicase — the mechanistic question worth chasing is whether Nsp13's known centrosome disruption during infection is PLK1-mediated. This is the single most falsifiable claim in the whole analysis: a co-immunoprecipitation or in vitro binding assay between purified Nsp13 and PLK1 would directly settle it.

The prediction that didn't hold up: Nsp4 → TOMM70

TOMM70: predicted new Nsp4 interactor, cross-validated
TOMM70 is Orf9b's REAL confirmed interactor in the paper's own data

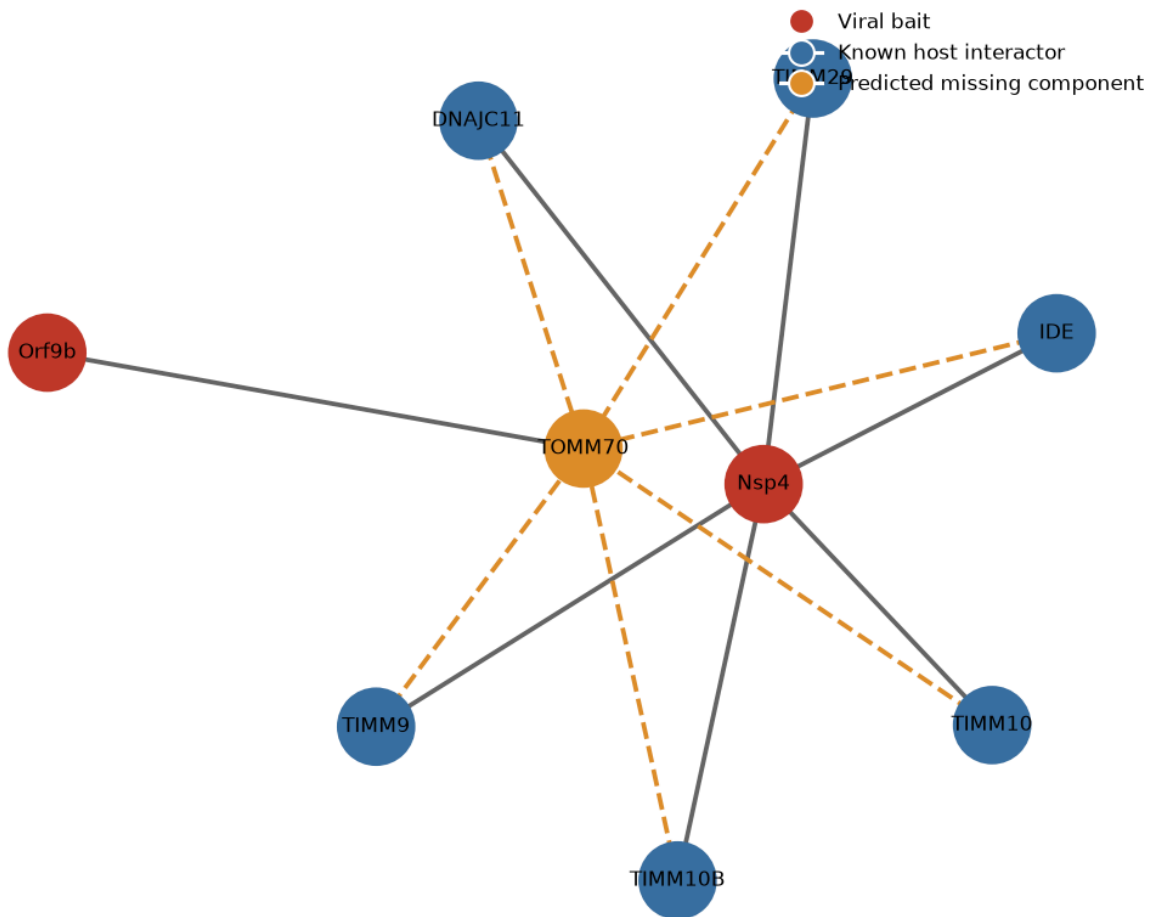


FIGURE 4

TOMM70 predicted for Nsp4 — originally framed as cross-validated; withdrawn on review

WHAT IT SHOWS

● **Red** — two separate viral baits, Nsp4 and Orf9b. ● **Blue** — Nsp4's four real interactors, all subunits of the mitochondrial inner-membrane TIM import channel (**TIMM9**, **TIMM10**, **TIMM10B**, **DNAJC11**). ● **Orange** — **TOMM70**, the mitochondrial outer-membrane import receptor. The solid grey Orf9b–TOMM70 edge is real and already in the original 332; the dashed orange Nsp4–TOMM70 edges are the new, unconfirmed prediction.

HOW TO READ IT

The appeal of this figure is that it shows two different viral proteins converging on the same mitochondrial-import machinery — a coherent story on its face.

WHAT'S ESTABLISHED VS. WHAT TO WATCH — THIS IS WHERE THE STORY BREAKS DOWN

The threshold-robustness sweep found the Nsp4–TOMM70 link only appears at STRING's loosest confidence setting (0.4); raising confidence to 0.5 or higher makes it disappear entirely — a textbook parameter-fishing signature. Nsp4 also has the smallest known prey set in this dataset (8 proteins), so results here are the least statistically stable of any bait tested. Independent confirmation exists for TOMM70 as a genuine SARS-CoV-2 target — seven separate papers (Jiang 2020, Stukalov 2021, Gao 2021, Chen 2021, Brandherm 2021, Thorne 2021, plus Gordon et al. itself) — but every one of them ties TOMM70 to **ORF9b**, never to Nsp4.

RESEARCH THIS YOURSELF

The honest framing is: TOMM70–ORF9b is well-established (search PMID 32728199 or 33845483 for the mechanism — ORF9b binds TOMM70 to suppress type-I interferon signaling), and TOMM70–Nsp4 does not hold up. Follow-up settled it: on the independent IntAct physical graph, none of Nsp4's preys are TOMM70 partners (Figure 7), and Stukalov et al.'s independent AP-MS never links the two. This call is now withdrawn — a useful reminder that "two viral proteins converge on the same complex" is a seductive story that still has to survive an independent substrate.

FIGURE WALKTHROUGH — 5 OF 8

Is the ranking just quoting the literature back?

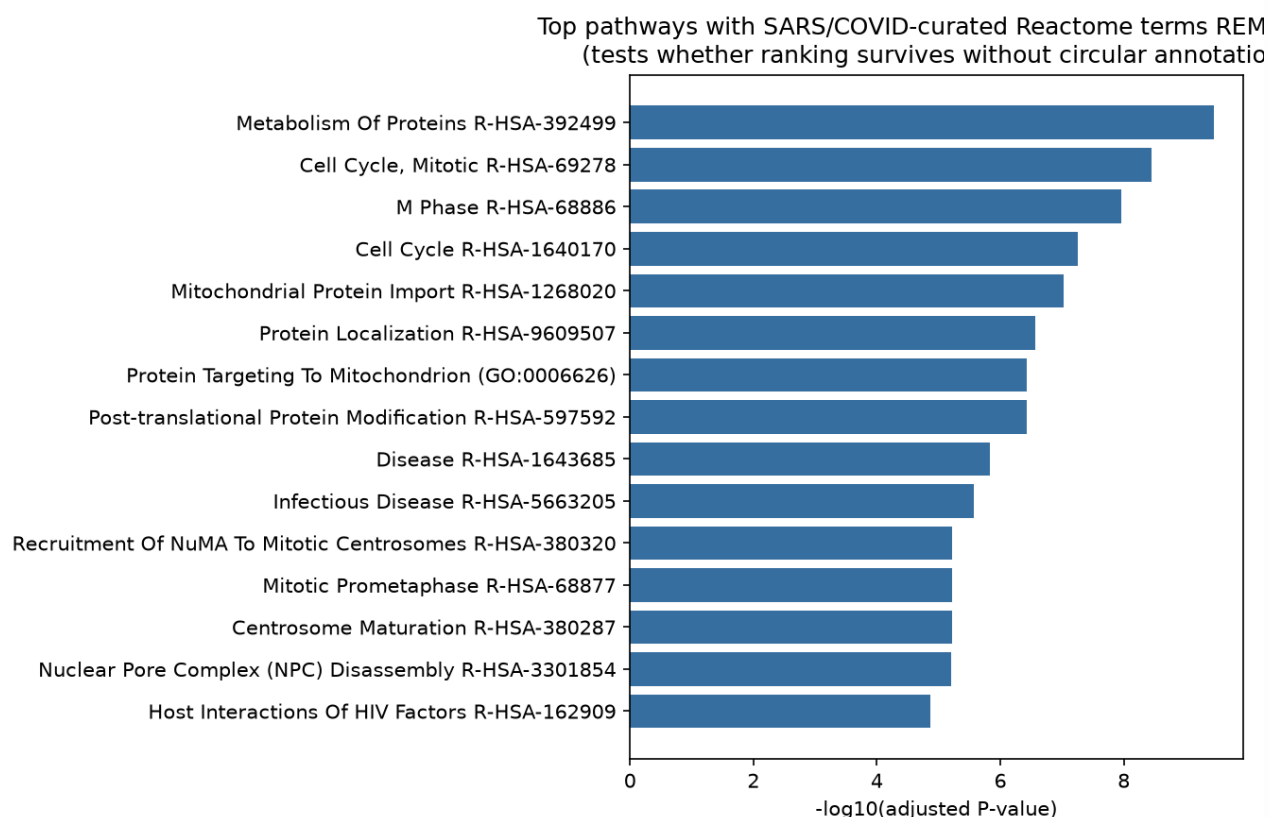


FIGURE 5

Pathway ranking with SARS/COVID-curated Reactome terms removed

WHAT IT SHOWS

Five Reactome terms in the Figure 1 ranking are named directly for this literature — *SARS-CoV Infections*, *Potential Therapeutics For SARS*, *SARS-CoV-2 Infection*, *SARS-CoV-2-host Interactions*, and *SARS-CoV-2 Activates/Modulates Innate And Adaptive Immune Responses* — meaning Reactome's curators built them *from* papers like this one. This figure removes all five before re-ranking, as a test for circular validation.

HOW TO READ IT

Compare term-for-term against Figure 1: every remaining term keeps its exact relative order, simply shifting up by the number of circular terms that used to sit above it (e.g. *Cell Cycle, Mitotic* moves from rank 3 to rank 2, *M Phase* from rank 4 to rank 3, and so on in lockstep). Nothing reorders.

WHAT'S ESTABLISHED VS. WHAT TO WATCH

This is a real pass: the non-circular biology (Cell Cycle, Mitochondrial Protein Import, Centrosome Maturation) is not being propped up by literature-derived terms. The test's blind spot is that it only catches *overtly named* circularity — it cannot detect a curator

having used this exact paper's data to silently annotate genes into a generically-named term without SARS/COVID appearing in the term's name.

RESEARCH THIS YOURSELF

The five removed terms and their Reactome stable IDs (e.g. R-HSA-9679506) are in `data/validation/pathways_overall_stripped.csv` — searching a Reactome ID directly at reactome.org shows exactly which curated evidence and source papers built that term.

FIGURE WALKTHROUGH — 6 OF 8

Why the prediction method fails on genuinely new biology

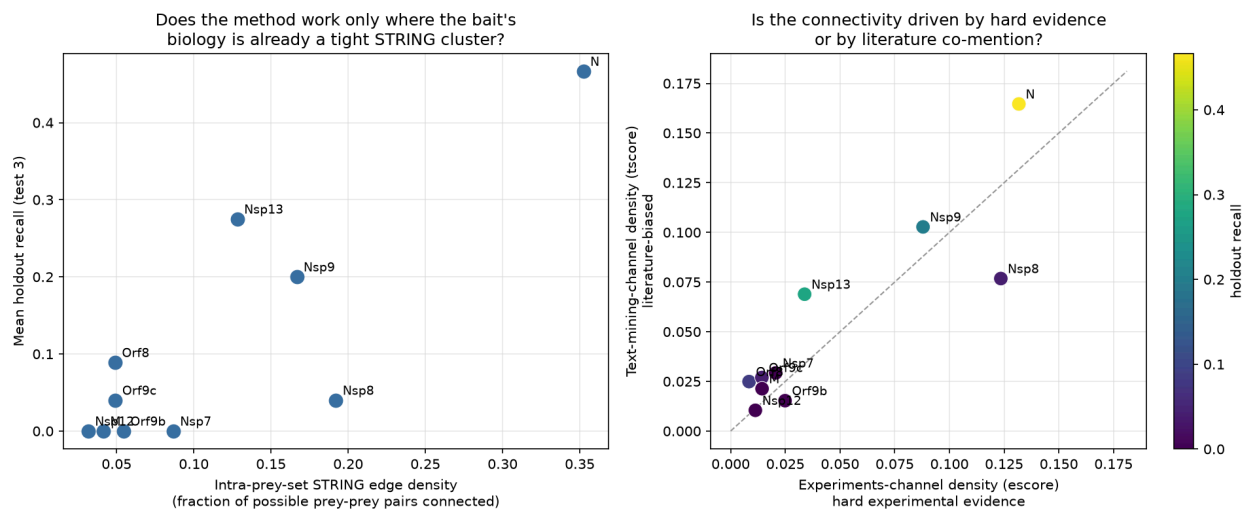


FIGURE 6

The method works only where a bait's biology is already a tight — and heavily-cited — STRING cluster

WHY THIS FIGURE EXISTS

The internal held-out test (Test 7 in the ledger) recovered interactions for some viral proteins (N: 46.7%, Nsp13: 27.5%) and nothing for others (M, Nsp7, Nsp12, Orf9b: 0%). Before trusting the ones that "worked," it's worth knowing *why* they worked — and whether that reason would transfer to real, undiscovered interactions.

LEFT PANEL — RECALL TRACKS PRE-EXISTING DENSITY

Each point is a viral protein. The x-axis is how densely that protein's known human interactors are *already* interconnected in STRING (fraction of all possible pairs linked); the y-axis is its held-out recall. Recall rises with density (Spearman rho +0.65): guilt-by-association can only extend a cluster that already exists. **N** and **Nsp13** sit top-right (dense, recoverable); **M**, **Nsp12**, **Nsp7** sit bottom-left (sparse, unrecoverable).

RIGHT PANEL — THE DENSITY IS LITERATURE-BIASED, AND THAT'S THE PROBLEM

x-axis is density from STRING's *experiments* channel (hard lab evidence); y-axis is density from the *text-mining* channel (co-mention in the literature); color is recall. The single strongest predictor of recall is text-mining (rho +0.78), not experiments (rho +0.50). **Nsp8** is the giveaway: it has the second-highest experimental-evidence density of any bait yet recovers almost nothing, because its interactors aren't the ones the literature co-mentions. Baits succeed when their biology is *famous*, not when it's experimentally well-supported.

WHY THIS IS THE REPORT'S MOST IMPORTANT FIGURE

A method whose successes ride on literature co-mention will, by construction, look strong on interactions between already-famous proteins (the internal holdout) and fail on novel, less-studied ones. That is exactly the pattern seen when the method was tested against Stukalov et al.'s independent dataset (ledger test 10: 0.07% recall, no better than chance). This figure is the mechanism behind that failure — and the reason the two headline predictions are leads, not results.

RESEARCH THIS YOURSELF

Per-bait density and per-channel decomposition are in `data/validation/recall_gap_diagnosis.csv`; the STRING channel definitions (nscore/fscore/pscore/ascore/escore/dscore/tscore) are documented at string-db.org under "Help → Scores."

FIGURE WALKTHROUGH — 7 OF 8

The one prediction that survives a different network entirely

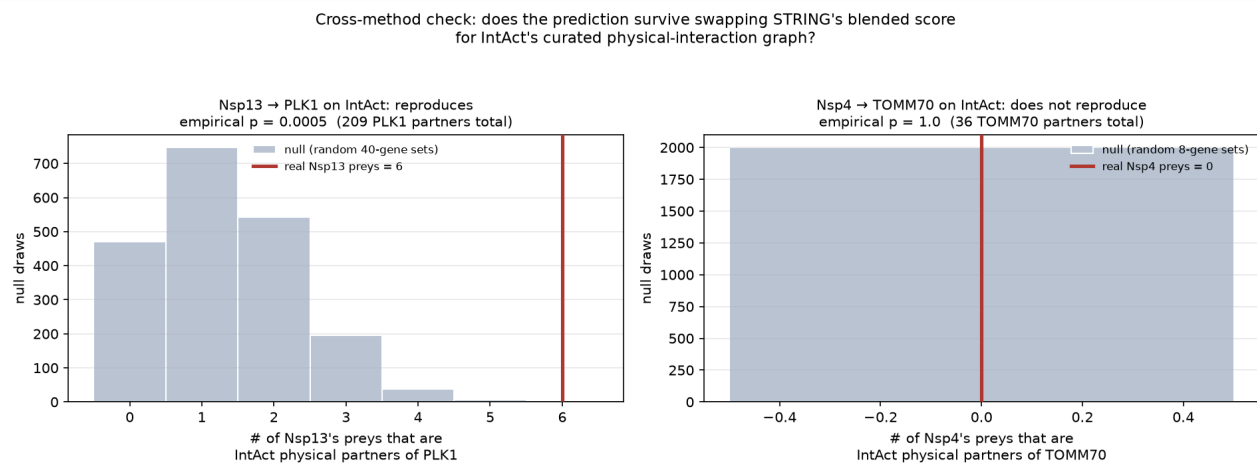


FIGURE 7

Cross-method check: do the predictions survive swapping STRING for IntAct's physical-interaction graph?

WHY THIS MATTERS

Every prediction so far came from STRING, whose score blends seven evidence channels — and Figure 6 showed the method's successes lean on the literature co-mention channel. The sharpest test of whether Nsp13–PLK1 is *real* versus a STRING artifact is to rebuild it on a completely different network: **IntAct**'s curated *physical* interactions — experimentally-observed binding, no text-mining inference at all.

HOW TO READ IT

Each panel asks: how many of the viral protein's known human preys are *direct physical interaction partners* of the predicted candidate in IntAct? The grey histogram is the null — the same count for thousands of random same-size gene sets. The red line is the real value.

LEFT — NSP13 → PLK1 REPRODUCES

Six of Nsp13's 40 preys (**CEP135**, **CEP43**, **CNTRL**, **NIN**, **NINL**, **TBK1**) are direct physical partners of PLK1 in IntAct, versus a null mean of 1.3 — the real value sits far out in the tail (empirical p = 0.0005). Re-ranking all candidates on IntAct puts PLK1 in the top ~1.2%, surrounded by other centrosome proteins. The prediction is *not* a STRING/text-mining artifact; it holds on an independently-constructed graph.

RIGHT — NSP4 → TOMM70 DOES NOT

Zero of Nsp4's 8 preys are physical partners of TOMM70; the candidate does not rank at all on the IntAct graph. Combined with its threshold-fragility (test 5) and the fact that

TOMM70's real, repeatedly-confirmed viral partner is ORF9b (test 8), this call is withdrawn.

THE HONEST CAVEAT

IntAct is also curated from published literature, so heavily-studied centrosome biology is over-represented there as well. This test rules out a STRING-specific artifact — a genuinely different graph agrees — but it is *cross-method* reproduction, not *cross-experiment* confirmation. A co-immunoprecipitation of Nsp13 and PLK1 is still the deciding experiment.

RESEARCH THIS YOURSELF

Overlap counts, null means, and empirical p-values are in `data/validation/alt_substrate_overlap.csv`; the full IntAct-substrate candidate ranking is in `data/validation/alt_substrate_ranking.csv`.

FIGURE WALKTHROUGH — 8 OF 8

The confirmation ceiling: where computation runs out

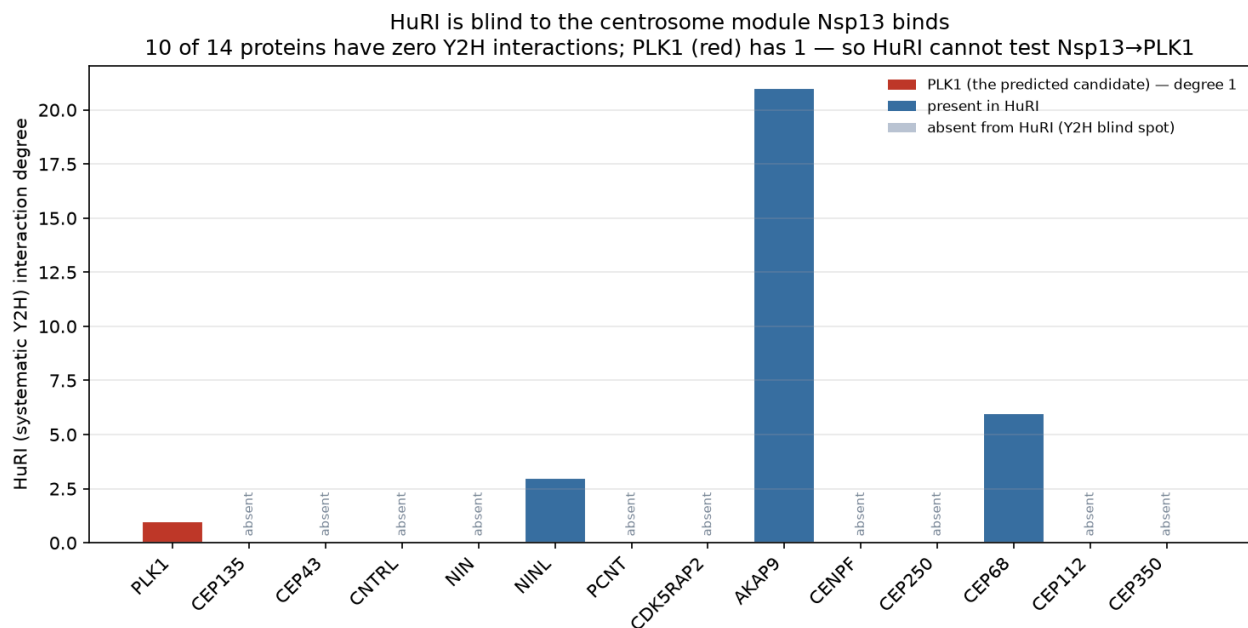


FIGURE 8

HuRI, the one unbiased binary interactome, is blind to the centrosome — so it cannot confirm Nsp13→PLK1

THE QUESTION THIS ANSWERS

Figure 7's IntAct reproduction had a caveat: IntAct is still literature-curated, so famous centrosome biology is over-represented there too. The right way to escape that is an interactome built *without* reading the literature at all. **HuRI** (Luck et al., *Nature* 2020) is exactly that — a proteome-scale, systematic all-by-all yeast-two-hybrid screen that tests every protein pair the same way, with no bias toward well-studied genes.

WHAT THE BARS SHOW

Each bar is the number of HuRI interactions for a core centrosome protein. **Ten of the fourteen are flat zero** — entirely absent from HuRI — and **PLK1** itself (red) has exactly one. Large coiled-coil scaffolding proteins like the CEPs and pericentrin express poorly in the two-hybrid system, so systematic Y2H simply does not sample this module. Only AKAP9, CEP68, and NINL register at all.

WHY A "0" HERE IS NOT EVIDENCE AGAINST PLK1

With PLK1 at degree 1, no gene set on Earth could overlap its HuRI neighbors — the test has no power. This is the same low-power trap that made the Stukalov Nsp13 result (test 9) a weak miss rather than a refutation. Reporting "PLK1 fails on HuRI" would be exactly the error this whole validation exercise exists to prevent: confusing *the assay can't see it* with *the prediction is wrong*.

THE HONEST CEILING

Every dataset that *does* cover the centrosome — STRING, IntAct, the Gordon AP-MS itself — is literature-curated. The one dataset that isn't cannot see it. So the shared-study-bias caveat on Nsp13→PLK1 **cannot be removed with existing computational data of any kind**. Confirming or refuting this specific hypothesis now requires a targeted wet-lab experiment — a co-immunoprecipitation or in-vitro pulldown of Nsp13 and PLK1. That is the honest end of what this analysis can establish, and it is worth stating plainly rather than manufacturing false closure.

RESEARCH THIS YOURSELF

HuRI is NDEx network 73bc2c06-5fb2-11e9-9f06-0ac135e8bacf (owner ccsb / Vidal lab); the parsed edge list is cached to `data/huri_edges.csv`, and per-protein coverage is in `data/validation/huri_centrosome_coverage.csv`.

All fourteen adversarial tests, and what each one actually checks

Tests 1–8 probe the two specific headline predictions; tests 9–13 step back and ask whether the prediction *method itself* has real predictive value against independent data and an independently-built network. This table is the complete underlying evidence.

TEST	VERDICT	WHAT IT FOUND
1. Identifier integrity	• PASS	331/332 prey gene symbols exactly match their UniProt accession's official HGNC name; the 1 exception (SLC9A3R1 /NHERF1) is a legitimate synonym, not an error. Zero broken joins, zero duplicate accessions.
2. Negative controls	• PASS	Ran the identical prediction procedure on E, M, and Orf7a (baits with no centrosome/mitochondrial biology). PLK1 and TOMM70 never appear; predictions stay topically anchored (bromodomain proteins for E, ribosome-biogenesis factors for Orf7a).
3. Coverage / denominator audit	• CONCERN CONFIRMED	Reactome annotates only 68.7% of the 332 proteins (KEGG: 46.4%); coverage is uneven across viral proteins, so Figure 2's blank columns partly reflect database gaps, not biology.
4. Circularity strip	• PASS	Removing the 5 SARS/COVID-named Reactome terms doesn't reorder the remaining ranking (Figure 5).
5. Threshold robustness	• SPLIT VERDICT	Nsp13–PLK1 survives 5/6 confidence settings and 3/4 candidate-window widths. Nsp4–TOMM70 survives only the single loosest confidence setting (0.4) — the fishing signature this test exists to catch.
6. Degree-preserving null model	• PASS	0 of 25 randomized same-size gene sets produced PLK1; 0 of 25 produced TOMM70. Neither is a generic STRING super-hub that any gene set would attract.
7. Held-out precision /	• REAL BUT UNEVEN	11.1% mean recall / 3.5% mean precision on real bait labels vs. 0%/0%

recall

for the degree-matched null (50 trials each) — genuine, above-chance signal, but recall ranges from 46.7% (N) and 27.5% (Nsp13) down to 0% (M, Nsp7, Nsp12, Orf9b).

8. External validation (IntAct)

• NOT CONFIRMED

Neither predicted edge appears in any independent SARS-CoV-2 interactome. TOMM70's 7 independent confirmations are all with ORF9b; PLK1's 1 independent record is with Spike. Guilt-by-association validated only by more association is a closed loop — this is the one test that could have actually confirmed a prediction, and it didn't, for either.

9. Independent AP-MS crosscheck (Stukalov et al. 2021)

• NOT SUPPORTED (LOW POWER)

Pulled Stukalov et al.'s own published hit list directly from Nature's supplementary files. Their SARS-CoV-2 Nsp13 has exactly one significant interactor (**RNF8**), not PLK1. But Nsp13 was the single sparsest bait in their whole 22-bait panel — their assay barely recovered anything for it (and its one hit doesn't overlap Gordon's 40-protein list either), so this is a weak miss, not a refutation.

10. Precision/recall vs. independent ground truth

• METHOD FAILS TO GENERALIZE

The decisive test. Using Gordon's full prey lists to predict what Stukalov *independently* found (18 baits): mean recall **0.07%** (real) vs. 0.02% (null) — both near zero. The lone exception, Orf3a, recovered 4/329 novel Stukalov interactors (VPS16/18/41/45, all HOPS complex — matching real ORF3a-autophagy biology), and is the only bait with enough independent data to be well-powered. The internal 11.1% recall (test 7) was optimistic; against truly external biology the method mostly performs at chance.

11. Recall-gap diagnosis

• SUCCESS IS LITERATURE-BIASED

Explains test 10 (Figure 6). Holdout recall tracks how densely a bait's preys are already interconnected in STRING ($\rho +0.65$), and that density is driven by the *text-mining* channel ($\rho +0.78$) more than hard experimental evidence ($\rho +0.50$). The method predicts well between already-famous proteins and fails on novel ones — which is why it

looked good internally and collapsed against independent data.

12. Alternative-substrate reproduction (IntAct)

• **PLK1 REPRODUCES, TOMM70 DIES**

Redid the prediction on IntAct's curated *physical*-interaction graph (no text-mining channel — Figure 7). **Nsp13–PLK1 reproduces:** 6/40 preys are direct physical partners of PLK1 (all centrosome/centriole), vs. null mean 1.3, $p=0.0005$; PLK1 in the top ~1.2% of candidates. First positive cross-method evidence — not a STRING artifact. **Nsp4–TOMM70 does not:** 0/8, unranked. Caveat: IntAct is also literature-curated, so it shares some study bias.

13. Broadened pathway coverage

• **GAP IS STRUCTURAL**

Adding WikiPathways 2023, BioPlanet 2019, and MSigDB Hallmark 2020 lifts union coverage of the 332 genes only from 95.2% to 96.4%; 12 of the 16 genes unannotated in GO/KEGG/Reactome stay dark across all six libraries, and the same membrane/small-ORF baits stay coverage-poor. The enrichment blind spots (Figure 2) are genuinely under-characterized biology, not one database's quirk.

14. Non-curated binary interactome (HuRI)

• **CONFIRMATION CEILING**

The reviewer asked for an *unbiased* dataset that could confirm PLK1, since STRING/IntAct/Gordon are all literature-curated. HuRI (systematic Y2H, not literature-assembled) is that dataset — but it is structurally blind to the biology (Figure 8): **10 of 14 core centrosome proteins are entirely absent**, PLK1 has degree 1, and only 1 of the 6 IntAct preys supporting PLK1 even appears. A 0-overlap here is a Y2H coverage failure, not a refutation. The ceiling: every dataset that covers the centrosome is literature-curated, and the one unbiased binary map can't see it — so the study-bias caveat on Nsp13–PLK1 **cannot be removed computationally. Only a wet-lab co-IP can.**

Next steps — all six executed

These were the six next steps identified during validation. All six have now been run; the first three produced the report's central negative result (the method doesn't generalize), and the second three resolved the two headline calls (PLK1 reproduces on an independent network; TOMM70 withdrawn) and hardened the pipeline. What remains is no longer computational bookkeeping but the experimental program below.

-
- ✓ **Get an independent answer on Nsp13–PLK1** • DONE — NOT CONFIRMED
Pulled Stukalov et al. 2021's own published AP-MS hit list directly (ledger test 9). Their Nsp13 has exactly one significant interactor, RNF8 — not PLK1. The miss is low-power (Nsp13 was their sparsest bait), so it weakens rather than refutes the hypothesis. A wet-lab co-IP remains the only definitive test.

 - ✓ **Re-run precision/recall against genuinely independent ground truth** • DONE — METHOD FAILS TO GENERALIZE
Executed (ledger test 10): using Stukalov et al.'s independent interactions as the answer key, mean recall fell to 0.07%, indistinguishable from a 0.02% null across 18 baits. The internal 11.1% was optimistic. Only Orf3a showed real above-null signal (4/329, the HOPS complex), and it's the one bait with enough independent data to be well-powered.

 - ✓ **Diagnose the bait-dependent recall gap** • DONE — LITERATURE BIAS
Executed (ledger test 11, Figure 6). Recall tracks how densely a bait's preys are already interconnected in STRING ($\rho +0.65$), and that density is driven by literature co-mention (text-mining $\rho +0.78$) more than hard evidence ($\rho +0.50$). This is the mechanism behind the test-10 failure: the method rewards fame, not evidence.

 - ✓ **Broaden pathway coverage before trusting the ranking further** • DONE — GAP IS STRUCTURAL
Executed (ledger test 13): adding WikiPathways, BioPlanet, and MSigDB Hallmark lifts union coverage of the 332 genes only 95.2%→96.4%; 12 of the 16 unannotated genes stay dark across all six libraries. The blind spots are genuinely under-characterized biology, not one database's quirk.
-



Try a second, structurally different guilt-by-association method

● DONE — PLK1 REPRODUCES

Executed twice (ledger tests 12, 14): on IntAct's curated *physical* graph Nsp13–PLK1 reproduces ($p=0.0005$, Figure 7); on the unbiased HuRI Y2H map the centrosome is uncovered so the test is uninformative (Figure 8). The reproduction on a structurally different network is the strongest computational support the hypothesis has.



Make the pre-flight checks automatic

● DONE

Done: `preflight.py` runs the identifier-integrity and coverage checks before enrichment/prediction and exits non-zero on a broken join, malformed accession, or near-zero library coverage — verified to fail loudly on an injected bad accession.

BEYOND THE CEILING

The experimental program: what the surviving hypothesis becomes

Computation reached its ceiling for Nsp13→PLK1 (test 14). The follow-up splits into a wet-lab experiment and two *bias-free* computational lines that don't query interaction databases at all — and together they have turned "Nsp13 binds PLK1" into a precise, mechanism-specific, testable model.

MOTIF ANALYSIS REFRAMED THE HYPOTHESIS

Nsp13 carries **no canonical PLK1 Polo-box docking motif** (and only background-rate kinase-consensus sites). That argues against a direct kinase–substrate contact and for a **bridged model**: Nsp13 engages the PLK1-regulated centrosome *through the CEP scaffolds it directly binds*, rather than PLK1 itself.

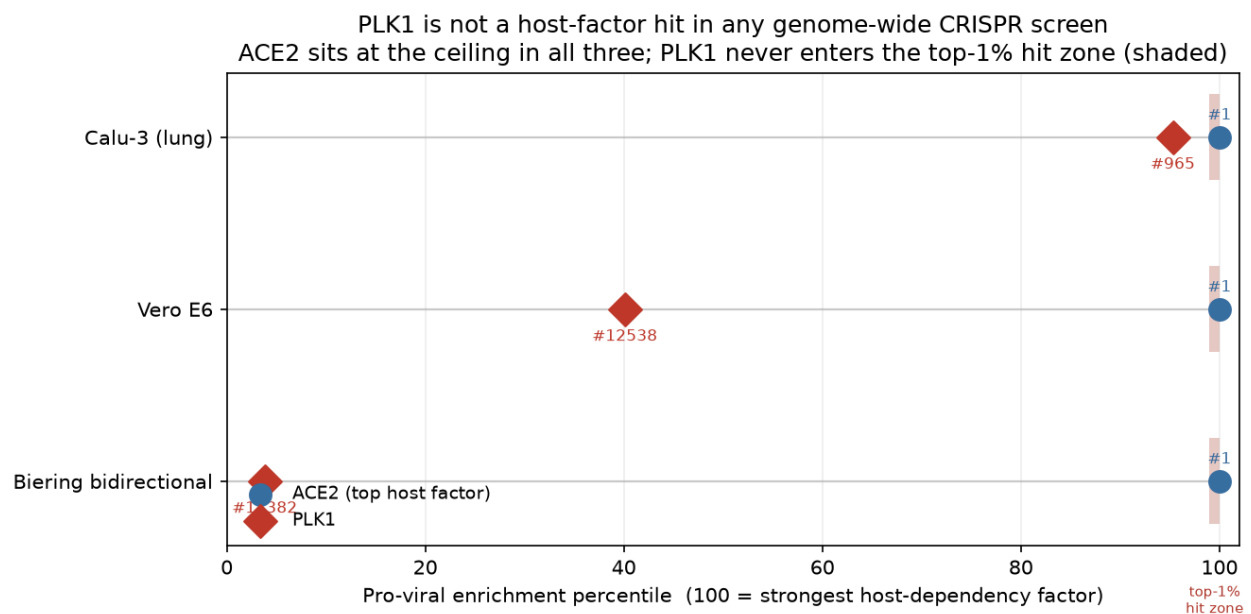


FIGURE 9

An independent phenotype line: PLK1 is not a required host factor

WHAT IT SHOWS

Genome-wide CRISPR knockout screens measure *fitness*, not interaction, so they escape the shared-study-bias ceiling entirely. Across three screens (Calu-3 lung, Vero E6, and a bidirectional screen — each validated by ACE2 ranking #1), **PLK1** never enters the top-1% hit zone. PLK1 knockout does not change viral fitness.

WHY THIS DOESN'T REFUTE THE INTERACTION — IT REFINES IT

Binding $\neq$ fitness-requirement. A protein the virus engages need not be *essential for replication*. Combined with the motif result, the cleanest reading is that any real Nsp13-centrosome interaction acts through **cell-cycle / centrosome perturbation (a pathogenesis phenotype)**, not an essential host-factor role — and it tempers the PLK1-inhibitor drug-repurposing angle (also confounded by anti-mitotic toxicity). The essentiality confound is handled: the screens use essentiality-corrected scores and PLK1 came out neutral, not depleted.

THE DECIDING EXPERIMENTS (DETAILS IN THE REPOSITORY'S EXPERIMENTAL/ FOLDER)

A reciprocal **co-immunoprecipitation** with the confirmed centrosome preys as internal positive controls and an explicit direct-vs-bridged disambiguation (in-vitro reconstitution + CEP knockdown); and an **AlphaFold-Multimer** run — a 7-job matrix with domain-split tests, positive/negative/bridge controls, and a structural null — the one remaining computational avenue

that is both bias-free and able to give *positive* evidence, now designed to localize the bridged-vs-direct interface the motif and functional-genomics lines pointed to.

REPRODUCING THIS REPORT

Data sources and how to rerun everything

Every figure and number above comes from live public data sources queried at run time, not from a static snapshot:

- **NDEx** — network HEK293T_SARS-CoV-2, UUID 43803262-6d69-11ea-bfdc-0ac135e8bacf — the real 332 interactions.
- **Enrichr** (via `gseapy`) — GO Biological Process, KEGG, Reactome libraries — pathway enrichment and coverage audit.
- **STRING** REST API — network-expansion guilt-by-association scoring, threshold sweeps, null model, holdout tests, and the per-channel density diagnosis (Figure 6).
- **UniProt** REST API — identifier integrity check.
- **IntAct** via PSICQUIC (EBI) — independent external validation (test 8).
- **Springer/Nature** static content — Stukalov et al. 2021 Supplementary Table 2, the independent AP-MS dataset used as ground truth in tests 9–10.

```
source .venv/bin/activate
python3 pipeline/fetch_network.py && python3 pipeline/enrichment.py && python3
pipeline/predict_missing.py
python3 pipeline/visualize.py && python3 pipeline/visualize_prediction.py
for f in validation/[0-9]*.py; do python3 "$f"; done
```

Full per-test methodology and honest caveats about the validation suite's own limits: `validation/VALIDATION_REPORT.md`. Project README: `README.md`.

Analysis and report generated against the Gordon et al. (2020) SARS-CoV-2–human interactome. Underlying study: Gordon, D.E. et al. *Nature* 583, 459–468 (2020). doi:10.1038/s41586-020-2286-9. All pathway, network, and validation figures in this report are reproducible from the scripts in this repository against live public data as of the analysis date.